

HUANG, Xinyang

Personal website: <https://huangxy-minel.github.io/>

Github: <https://github.com/Huangxy-Minel>

Email: xhuangci@connect.ust.hk

Mobile: +86-150-2236-3025

EDUCATION

- Hong Kong University of Science and Technology** Hong Kong, China
Doctor of Philosophy - Computer Science & Engineering Aug. 2023 - Dec. 2026 (Expected)
- Hong Kong University of Science and Technology** Hong Kong, China
Master of Philosophy - Computer Science & Engineering; GPA: 3.73/4.0 Sept. 2021 - Aug. 2023
Courses: Advanced Algorithms (A-), Computer Networks (A-), Machine Learning (A-), Advanced Computer Architecture (A+)
- University of Electronic Science and Technology of China** Chengdu, China
Bachelor of Science in Network Engineering; GPA: 3.88/4.0 Sept. 2017 - June 2021
Courses: Graph Theory, Stochastic Processes, TCP/IP Protocol, Access Network, Signal and Systems, Digital Circuits, etc.

SKILLS SUMMARY

Languages: C/C++, Python, CUDA, MATLAB, Verilog, Java, etc.
SDKs/Libraries: DPDK, DOCA, NCCL, NVSHMEM, libibverbs, DeepGEMM, DeepEP, eBPF/XDP, etc.
Platforms/Frameworks: BlueField, ConnectX, Corundum, PyTorch, vLLM, SGLang, Spark, etc.
Tools: Docker, perf, Nsight, Vivado/Vitis, cocotb, MAAS, Keil, etc.

SELECTED PUBLICATIONS

- Tuning Host Datapath Performance with PipeTune:** *1st author*, NSDI'27
- CEIO: A Cache-Efficient Network I/O Architecture for NIC-CPU Data Paths:** *co-1st author*, SIGCOMM'25
- Enabling Efficient GPU Communication over Multiple NICs with FuseLink:** *4th author*, OSDI'25
- Cache-Aware Rate Control for RDMA I/O Congestion:** *2nd author*, APNet'25
- Accelerating Privacy-Preserving Machine Learning with GeniBatch:** *1st author*, EuroSys'24
Under Review:
- EtherNode: Scaling Up GPU Nodes over Commodity Ethernet:** *1st author*
- XDD: eXpressed DPU Data-Plane Runtime:** *1st author*
- FreeIO: Enabling Zero-Shuffle and Sub-RTT GPU Communication for MoE Inference:** *co-1st author*
- Enabling HBM Pooling over Scale-Up Fabric for KV-Cache Storage:** *2nd author*

INTERNSHIP

- SuperPod Architecture Design and Performance Optimizations (Project Up)** Tencent, Shenzhen
Researcher, Director: Dr. Yinben XIA and Xiang LI Sept 2025 - Present
 - Tech:** RoCE, NCCL, CUDA, DeepGEMM, DeepEP, Optics, Multi-Path Transport.
 - EtherNode:** Co-design and deploy Tencent's next-generation GPU SuperPod that scales up to 512 GPUs with 800GB/s per-GPU bisection bandwidth. The interconnects are based on commodity Ethernet and expose a CCL-level middleware for multi-path RDMA collective communication.
 - Scale-Up RAM:** Co-design and deploy Tencent's high-speed RAM tray over scale-up fabric, which provides 2 TB memory extension for GPU HBM and accelerates long-context inference by up to 4.5× for recent models such as GLM-5.2 and DeepSeek-v3.2.

SELECTED RESEARCH EXPERIENCE

- Optimizations Towards 100Gbps+ Host Networks** HKUST, Hong Kong
Researcher, Supervisor: Professor Kai CHEN Sept. 2023 - Present
 - Tech:** DPDK, DOCA, BlueField, rdma-core.
 - CEIO:** Design a cache-efficient I/O architecture based on the latest NVIDIA BlueField 3 DPUs, introducing proactive, credit-based I/O rate control and elastic buffering to eliminate LLC misses in I/O datapaths.
 - * **Results and Progress:** CEIO outperforms SOTA solutions by up to 2.9× in throughput and 1.9× in latency. CEIO has been accepted by SIGCOMM'25 (co-1st author).
 - PipeTune:** Build a systematic tuning framework that models the correlation among 6 datapath configurations and memory-subsystem efficiency, enabling bootstrap-time optimization and runtime diagnosis for 100Gbps+ datapaths.
 - * **Results and Progress:** Our framework improves the throughput and reduces P99.9 latency of target datapaths (i.e., eRPC, Open vSwitch, etc.) by up to 2.1× and 4.6×, respectively.
- Communication and Memory Optimizations for LLM Inference** HKUST, Hong Kong
Researcher, Supervisor: Professor Kai CHEN Jan. 2025 - Present
 - Tech:** CUDA, NCCL, NVSHMEM, DOCA, GPU Copy Engine.
 - FreeIO:** Design an arbiter-based MoE all-to-all collective that decouples correctness enforcement from data transfer, enabling direct-to-destination writes with zero data shuffling and sub-RTT coordination.
 - * **Results and Progress:** FreeIO reduces the MoE collective tax by up to 12× compared to widely-used libraries such as DeepEP and NCCL.

- **HBM Pooling:** Enable KV-cache storage over scale-up fabric by pooling remote HBM across GPUs, with adjacent page merging and batched copy engine submission to minimize transfer control overhead.

- * **Results and Progress:** Expand per-GPU KV-cache capacity beyond local HBM limits, improving inference batch size and system throughput.

- **High-Performance and Flexible DPU Infrastructure**

HKUST, Hong Kong

Nov. 2022 - Dec. 2025

- *Researcher, Supervisor: Professor Kai CHEN*

- **Tech:** DPDK, BlueField, DOCA, eBPF/XDP, Corundum, Vivado/Vitis, cocotb, Verilog.
- **XDD:** Extend existing DPU offload frameworks (i.e. NVIDIA DOCA) with efficient data movement, lightweight queue management, and elastic context switching.
- * **Results and Progress:** Complete XDD development with 20000+ LoC. Compared to DOCA, XDD improves various DPU workloads (e.g., NF chain, distributed protocols, RDMA, etc.) by up to 4.8× in throughput.
- **SingNIC:** Design a 100Gbps programmable NIC architecture with on-path MIPS cores.
- * **Results and Progress:** Build an FPGA prototype that can offload XDP programs with line rate.

- **Privacy-Preserving Machine Learning (PPML) System**

HKUST, Hong Kong

Dec. 2021 - Oct. 2022

- *Researcher, Supervisor: Professor Kai CHEN*

- **Tech:** Docker, FATE, Spark, HDFS, Python, CUDA.
- **GeniBatch:** Design a batch compiler called GeniBatch that translates a PPML program with Partial Homomorphic Encryption into an efficient program with batch optimization.
- **Results and Progress:** GeniBatch accelerates end-to-end performance for various cross-silo PPML applications from 1.59x to 22.6x. GeniBatch has been accepted by EuroSys'24 (1st author).

HONORS AND AWARDS

- Honorable Mention, Best Student Paper Award @ ACM SIGCOMM 2025
- Third Place, NVIDIA Hackathon China - 2025
- TDSC Reviewer Certificate - 2025
- Postgraduate Studentship (PGS) award of HKUST - 2021-2022, 2022-2023, 2023-2024, 2024-2025, 2025-2026
- Outstanding Academic Scholarship of UESTC for full 3 academic years - 2017-2018, 2018-2019, 2019-2020
- National Innovation and Entrepreneurship Excellent Project - 2018-2019
- Second prize in National Electronic Design Competition - Aug. 2019

TEACHING EXPERIENCE

- **Teaching Assistant**

HKUST, Hong Kong

- *Course: C++ Language, Operating System, Computer Architecture, etc.*

Feb. 2022 - Present

- Conducted tutorials, designed experiments and answered questions.

FULL PUBLICATION LIST

1. **Xinyang Huang**, Zhuobin Huang, Qijing Li, Junxue Zhang, Mengyu Ma, Han Tian, Bowen Liu, and Kai Chen. "PipeTune: An Efficient Performance Tuning Framework for Host Datapaths." In *USNIX NSDI 2027*.
2. Bowen Liu*, **Xinyang Huang***, Qijing Li, Zhuobin Huang, Yijun Sun, Wenxue Li, Junxue Zhang, Ping Yin, and Kai Chen. "CEIO: A Cache-Efficient Network I/O Architecture for NIC-CPU Data Paths." In *ACM SIGCOMM 2025*. (* co-first author)
3. Zhenghang Ren, Yuxuan Li, Zilong Wang, **Xinyang Huang**, Wenxue Li, Kaiqiang Xu, Xudong Liao, Yijun Sun, Bowen Liu, Han Tian, Junxue Zhang, Mingfei Wang, Zhizhen Zhong, Guyue Liu, Ying Zhang, and Kai Chen. "Enabling Efficient GPU Communication over Multiple NICs with FuseLink." In *USENIX OSDI 2025*.
4. Xudong Liao, Yijun Sun, Han Tian, Xinchun Wan, Yilun Jin, Zilong Wang, Zhenghang Ren, **Xinyang Huang**, Wenxue Li, Kin Fai Tse, Zhizhen Zhong, Guyue Liu, Ying Zhang, Xiaofeng Ye, Yiming Zhang, and Kai Chen. "MixNet: A Runtime Reconfigurable Optical-Electrical Fabric for Distributed Mixture-of-Experts Training." In *ACM SIGCOMM 2025*.
5. Wenxue Li, Xiangzhou Liu, Yunxuan Zhang, Zihao Wang, Wei Gu, Tao Qian, Gaoxiong Zeng, Shoushou Ren, **Xinyang Huang**, Zhenghang Ren, Bowen Liu, Junxue Zhang, Kai Chen, and Binyang Liu. "Revisiting RDMA Reliability for Lossy Fabrics." In *ACM SIGCOMM 2025*.
6. Qijing Li, **Xinyang Huang**, Bowen Liu, Pengbo Li, Junxue Zhang, and Kai Chen. "Cache-Aware I/O Rate Control for RDMA." In *ACM APNet 2025*.
7. Xinchun Wan, Luyang Li, Han Tian, Xudong Liao, **Xinyang Huang**, Chaoliang Zeng, Zilong Wang, Xinyu Yang, Ke Cheng, Qingsong Ning, Guyue Liu, Layong Luo, and Kai Chen. "A Generic and Efficient Communication Framework for Message-level In-Network Computing." In *IEEE INFOCOM 2025*.
8. **Xinyang Huang**, Junxue Zhang, Xiaodan Cheng, Hong Zhang, Yilun Jin, Shuihai Hu, Han Tian, and Kai Chen. "Accelerating Privacy-Preserving Machine Learning with GeniBatch." In *ACM EuroSys 2024*.